Troubleshooting Guide — Symptom Reference

Quick Reference · All Machine Types

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Owner	Maintenance Department
Scope	CNC Milling, Press Line, Assembly Line — all plants

1. Purpose

This guide is organized by observed symptom rather than by machine or error code, so operators and technicians can look up a problem the way it is actually reported on the floor. Each entry cross-references the relevant downtime code and the detailed SOP or reference document for full corrective steps.

2. Symptom → Cause → Action Table

Symptom	Possible Cause	Recommended Action
High spindle/bearing vibration	Bearing wear or damage	Stop machine, inspect bearing, replace per Bearing_Replacement_SOP
Slow cycle time / falling behind takt	Tool wear or incorrect feed/speed setting	Inspect and replace worn tooling; verify program feed/speed against sp
Frequent jams on conveyor or feed path	Sensor misalignment or part orientation issue	Inspect and realign sensor per Electrical_Troubleshooting_SOP (code D
Sudden loss of hydraulic pressure	Hose, seal, or fitting failure	Depressurize and inspect per Hydraulic_Maintenance_SOP (code DT-0
Hydraulic oil discoloration or foaming	Water contamination or oil breakdown	Sample oil, compare to Lubrication_Schedule spec; drain and refill if ou
Repeated short stoppages (a few minutes each)	Sensor or limit switch fault	Check PLC fault log for the flagged I/O address; clean/realign/replace s
Parts trending out of dimensional tolerance	Tool offset drift or fixture wear	Re-verify tool offsets/fixture per Calibration_SOP (code DT-06); do not
Torque tool failing calibration check	Clutch wear or tool drift	Remove from service immediately, tag 'Out of Calibration', send for reca
Unexpected E-stop / safety trip	Guarding interruption, light curtain break, or wiring fault	Do not bypass — inspect guarding and wiring per Electrical_Troublesho
Excessive changeover time	Internal (machine-stopped) setup steps not pre-staged	Review changeover sequence against Changeover_SOP (code DT-03)
Line waiting on incoming parts	Upstream material/feed shortage	Escalate to logistics per PM_Guide; check buffer stock levels (code DT-
Unusual noise from gearbox or drive	Lubrication breakdown or component wear	Check lubrication level/type against Lubrication_Schedule; inspect for w
Recurring downtime at shift start	Handover/briefing gap between shifts	Reinforce shift handover checklist per Plant_Maintenance_Policy (code
Quality defects rising without a clear machine fault	Calibration drift on inspection gauges/tools	Verify gauge calibration status per Calibration_SOP before assuming a
Coolant/oil leak pooling under machine	Seal or hose degradation	Isolate and inspect nearest hydraulic/coolant line per Hydraulic_Mainten

3. Escalation Reminder

If the recommended action does not resolve the symptom within one attempt, or the underlying downtime code is classified Major (DT-01, DT-02, DT-08) in the Error_Code_Handbook, escalate immediately per the Plant_Maintenance_Policy escalation matrix rather than repeating troubleshooting steps.

■ **This guide is a triage aid, not a substitute for the full SOP. Always follow lockout-tagout in Safety_Guidelines before physical inspection or repair.**